📘 Day 1: Introduction + Python Basics

Today we will start with:

1. Course Introduction
2. What is AI
3. Python Basics
4. Exercises

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1. Course Introduction

- You will learn Python → Machine Learning → Deep Learning → AI Applications.
- Be ready to code with me in every class.

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2. What is AI?

Artificial Intelligence (AI) is the ability of machines to mimic human intelligence.

Examples:

- Google Maps
- Netflix Recommendations
- Chatbots (like ChatGPT)
- Drones & Robotics
- Face Recognition

AI vs ML vs DL

- **AI**: Broad field, making machines think smartly.
- **ML**: Teaching machines from data.
- **DL**: Using neural networks for advanced problems.

In AI and Machine Learning, we mainly work with three types of learning:

Supervised Learning

when data is labeled

like predicting marks, prices, diseases

Unsupervised Learning

when data is unlabeled

like grouping customers

Reinforcement Learning

learning by reward and punishment

used in robotics, drones, games

NAVTTC AI/ML Course - Lab 1

Introduction to Python Basics

In this first lab, we will cover the fundamental concepts of Python that will be useful throughout the course.

Topics Covered:

- Tuples
- Dictionaries
- Loops
- Functions & Methods
- List Methods (`append` , `count` , `extend` , `insert` , `pop` , `remove` , `reverse` , `sort`)

In [ ]:
```
# Print & Comments
"Hello AI World!"
# This is a comment
```

In [5]:
```
# Variables & Data Types
name = "Asma"
age = 24
height = 5.4
is_student = True

print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
```

```
<class 'str'>
<class 'int'>
<class 'float'>
<class 'bool'>
```

List

List is an ordered, mutable collection that can store multiple items of different data types.

In [21]:
```
fruits = ["apple", "banana", "cherry"]
fruits.append("mango")

print(fruits)
print("First element:", fruits[0])
```

```
print("Last element:", fruits[-1])
print("Slice [1:3]:", fruits[1:3])
```

```
['apple', 'banana', 'cherry', 'mango']
First element: apple
Last element: mango
Slice [1:3]: ['banana', 'cherry']
```

List Methods

Python gives us built-in list methods to work faster

append() → add item

insert() → add at position

remove() → delete item

pop() → remove by index

sort() → arrange

reverse() → reverse order

count() → count items

extend() → merge lists

Tuples

A **tuple** is an ordered, immutable collection of elements.

In [9]:
```
colors = ("red", "blue", "green")
print(colors[0])
#colors[0] = "yellow"
```

```
red
```

Dictionary

A **dictionary** stores data in key-value pairs.

In [24]:
```
student = {"name": "Ali", "age": 21, "grade": "A"}
print(student["name"])
student["age"] = 22
print(student)
```

```
Ali
{'name': 'Ali', 'age': 22, 'grade': 'A'}
```

Loop

A loop is used to repeat a block of code multiple times without writing it again and again

for loop → best when you know how many times to repeat (or when looping through a list).

while loop → best when you want to repeat until a condition is met.

In [26]:
```python
# Loops
# For loop
for i in range(5):
    print("Iteration:", i)

# Loop through list
fruits = ["apple", "banana", "cherry"]

for fruit in fruits:
    print("I like", fruit)
```

```
Iteration: 0
Iteration: 1
Iteration: 2
Iteration: 3
Iteration: 4
I like apple
I like banana
I like cherry
```

In [27]:
```python
#For loop with condition
for num in range(1, 11):
    if num % 2 == 0:
        print(num, "is even")
```

```
2 is even
4 is even
6 is even
8 is even
10 is even
```

In [11]:
```python
password = ""
while password != "python":
    password = input("Enter the password: ")
print("Access Granted ✅")
```

```
Access Granted ✅
```

In [ ]:
```python
def is_prime(number):
    if number <= 1:
        return False
    for i in range(2, number):
        if number % i == 0:
            return False
    return True

# Take input from user
num = int(input("Enter a number: "))

# Check and print result
if is_prime(num):
    print(num, "is a Prime Number")
else:
    print(num, "is NOT a Prime Number")
```

In [ ]:
```python

```

4. Exercises

1. Create a list of 5 numbers and print their sum.
2. Create a dictionary for a car (brand, model, year) and print it.
3. Write a program that checks if a number is even or odd.
4. Create a password system where: The user gets maximum 3 attempts

If the password is correct → show Access Granted

If the password is wong → show Wrong password

After 3 wrong attempts → show Account Locked

- Practice list, tuple, dictionary operations.

5. Create a dictionary that stores marks of a student in:

Math

English

Science

Then:

Calculate total marks

Calculate average marks

In [ ]: